

Introduction to Data Management

PROJECT REPORT

(Project Semester January-April 2025)

Demographic Analysis of Border Crossing Entry Data in the United States: An Interactive
Border Crossing Dashboard Approach

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Program: B. Tech - Computer Science and Engineering

Course Code: INT217

Under the Guidance of

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Discipline of CSE/IT

Lovely School of Computer Science and Engineering

Lovely Professional University, Phagwara



CERTIFICATE

This is to certify that Rahil Khan bearing Registration no. 12309489 has completed INT217 (Introduction to Data Management) project titled, “Demographic Analysis of Border Crossing Entry Data in the United States: An Interactive Border Crossing Dashboard Approach” under my guidance and supervision. To the best of my knowledge, the present work is the result of his original development, effort and study.

Signature

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Designation of the Supervisor

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Date:11/04/2024

DECLARATION

I, Rahil Khan, student of B.Tech Computer Science and Engineering under CSE/IT Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

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A small, rectangular image showing a handwritten signature in black ink. The signature appears to be 'Rahil' with a long horizontal stroke extending to the right.

11/04/2025

Acknowledgement

The opportunity of attaining a course based on Data Management using Excel at Lovely Professional University was worth learning. It was a prestige for me to be part of it. During the period of my course, I received tremendous knowledge related to Microsoft Excel and Data Management.

Pre-eminently, I would like to express my deep gratitude and special thanks to my course teacher Jaffar Amin Chacket for her theoretical knowledge and encouragement on this project and for her valuable guidance and affection for the successful completion of this project.

Secondly, I would like to thank Lovely Professional University for giving me an opportunity to learn this course.

Lastly, I would like to thank the almighty and my parents for their constant encouragement, moral support, personal attention, and care.

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A small, rectangular image showing a handwritten signature in black ink. The signature appears to be 'Rahil' with a stylized flourish.

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Abstract

Excel is a software program created by Microsoft that uses spreadsheets to organize numbers and data with formulas and functions. Excel analysis is ubiquitous around the world and used by businesses of all sizes to perform data analysis. Excel features calculation, graphing tools, pivot tables, and a macro programming language called Visual Basic for Applications, and several other features which make Excel a perfect choice to manage and analyze data.

My project is an Excel Border Crossing Dashboard. The Excel Border Crossing Dashboard is used to display overviews of large data tracks. Excel Border Crossing Dashboards use dashboard elements like tables, charts, and gauges to show the overviews. The dashboards ease the decision-making process by showing the vital parts of the data in the same window.

In this report, I have shared a project where I have done data analysis of a Border Crossing dataset across US States. This report also presents my learning during my course classes.

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Chapter 1 - Introduction

I have created an Excel Border Crossing Dashboard of a Border Crossing dataset across US States. This dashboard explains and highlights important facts, trends, and patterns related to border crossing statistics in different states based on gender, race, age group, and year.

The dataset used contains information regarding border crossing counts in the United States. It includes details such as gender-wise data, race-wise data, Hispanic origin, and different age groups of people affected by border crossing. The dataset also covers records from multiple years across various US states.

I have scrubbed and organized the entire dataset and performed the analysis on a clean data sheet created within the Excel workbook. I have derived and calculated important results from the dataset using various Excel features like pivot tables, slicers, and functions. I have represented the findings in the form of a dynamic dashboard using Excel's visualization tools and various charts.

Chapter 2 - Objectives

This project on Border Crossing Analysis provides the records, facts, and trends of border crossing statistics across US States with respect to gender, race, Transportation mode, and year. However, here are the few main objectives that are discussed in the dashboard:

- Showing the total number of border crossing cases and their distribution with respect to gender, race, age group, year, and state.
- Showing the performance and comparison of states overall, or in a particular year, based on border-type specific border crossing statistics.
- Showing the distribution and trends of border crossing statistics among different age groups.
- Highlighting the gender-wise comparison of border crossing statistics overall, or in specific states, or particular years.
- Showing the trend analysis of border crossing statistics over the years based on different states and entry point groups.
- Displaying the most affected age groups and races in terms of border crossing statistics.
- Highlighting the state-wise contribution to the total number of border crossing cases.
- Visualizing the overall trends and patterns to provide meaningful insights for decision-making and awareness.

Chapter 3 - Source of Dataset

The dataset is taken from Data.gov. Data.gov is the official open data portal of the United States Government that provides access to a wide range of datasets for public use. This platform allows users to find and download datasets published by government agencies.

I have selected a Border Crossing dataset which contains important details of crossing rates across the United States categorized by gender, mode, age latitudes/ longitudes, and year.

Here are the details of my chosen dataset:

Name -Border Crossing Dataset— United States

Link -

<https://drive.google.com/file/d/1ZvzXLQENMmH3KhauACIWInQKfBecsZn4/view?usp=sharing>

Format – CSV

Data Fields:

- PORT NAME— (String) — Names of port
- STATES — (String) — States of United States
- PORT CODE — (Numeric) — Numerical code for unit type
- BORDER — (String) — Border Names
- STUB_NAME_NUM — (Numeric) — Numerical code for STUB_NAME
- DATE — (String) — Data of Dates
- MEASURES — (String) — Transportation Mode used
- VALUE — (Integer) — Code record
- LATITUDE— (Double) — Geo-Location
- LONGITUDE — (Double) — Geo - Location
- POINT — (Coordinates) — Geo - Location

RETURN — (String) -- Individual Return

Chapter 4 - Dataset Preprocessing

In this project, the dataset preprocessing was an important step to ensure accurate and meaningful analysis. The dataset was downloaded from the official United States Government Open Data Portal (Data.gov). The raw dataset contained information about border crossing crossing counts in the United States categorized by gender, race, transportation mode, and year.

The following preprocessing steps were performed:

1. Data Cleaning:
 - Removed unnecessary columns that were not required for analysis.
 - Identified and handled missing values in important columns.
 - Replaced blank cells or missing entries with appropriate values where applicable.
2. Data Transformation:
 - Renamed column headers for easy understanding and readability.
 - Organized data into a clean and structured format suitable for analysis.
 - Filtered relevant records and removed inconsistent data entries.
3. Creating Clean Dataset Sheet:
 - After cleaning and transforming the data, a new sheet named "Clean Dataset" was created within the Excel workbook.
 - The following screenshots represent:

Port Name	State	Port Code	Border	Date	Measure	Value	Latitude	Longitude	Point	Return
Jackman	Maine	104	US-Canada Bord	Jan-24	Trucks	6556	45.806	-70.397	POINT (-70.396722 45.806)	Yes
Porthill	Idaho	3308	US-Canada Bord	Apr-24	Trucks	98	49	-116.499	POINT (-116.49925 48.99)	No
San Luis	Arizona	2608	US-Mexico Bord	Apr-24	Buses	10	32.485	-114.782	POINT (-114.782222 32.485)	No
Willow Creek	Montana	3325	US-Canada Bord	Jan-24	Pedestrians	2	49	-109.731	POINT (-109.731333 48.9)	No
Warroad	Minnesota	3423	US-Canada Bord	Jan-24	Personal Vehicle Pass	9266	48.999	-95.377	POINT (-95.376555 48.99)	No
Whittash	Montana	3321	US-Canada Bord	Jan-24	Personal Vehicles	29	48.997	-111.258	POINT (-111.257916 48.99725)	Yes
Ysleta	Texas	2401	US-Mexico Bord	Jan-24	Personal Vehicle Pass	521714	31.673	-106.335	POINT (-106.335448 31.673126)	Yes
Warroad	Minnesota	3423	US-Canada Bord	Jan-24	Trucks	837	48.999	-95.377	POINT (-95.376555 48.99)	Yes
Wildhorse	Montana	3323	US-Canada Bord	Jan-24	Trucks	20	48.999	-110.215	POINT (-110.215083 48.99361)	Yes
Wildhorse	Montana	3323	US-Canada Bord	Jan-24	Personal Vehicle Pass	965	48.999	-110.215	POINT (-110.215083 48.99361)	No
Warroad	Minnesota	3423	US-Canada Bord	Jan-24	Bus Passengers	278	48.999	-95.377	POINT (-95.376555 48.99)	Yes
Westhope	North Dakota	3419	US-Canada Bord	Jan-24	Truck Containers Load	102	49	-101.017	POINT (-101.017277 48.99611)	No
Warroad	Minnesota	3423	US-Canada Bord	Jan-24	Truck Containers Load	459	48.999	-95.377	POINT (-95.376555 48.99)	Yes
Ysleta	Texas	2401	US-Mexico Bord	Jan-24	Truck Containers Empty	21355	31.673	-106.335	POINT (-106.335448 31.673126)	Yes
Sweetgrass	Montana	3310	US-Canada Bord	Jan-24	Truck Containers Empty	1985	48.998	-111.96	POINT (-111.959611 48.99388)	Yes
Brownsville	Texas	2301	US-Mexico Bord	Jan-24	Rail Containers Empty	7712	25.952	-97.401	POINT (-97.40067 25.9515)	Yes
Sumas	Washington	3009	US-Canada Bord	Jan-24	Buses	14	49.002	-122.265	POINT (-122.264805 49.002388)	Yes
Madawaska	Maine	109	US-Canada Bord	Jan-24	Truck Containers Empty	130	47.36	-68.329	POINT (-68.326884 47.360052)	No
Dalton Cache	Alaska	3106	US-Canada Bord	Jan-24	Truck Containers Load	21	59.451	-136.362	POINT (-136.361944 59.450556)	No
Calais East	California	2507	US-Mexico Bord	Jan-24	Personal Vehicle Pass	435768	32.673	-115.388	POINT (-115.388018 32.673389)	Yes
Pinetree	Minnesota	3425	US-Canada Bord	Jan-24	Personal Vehicle Pass	188	49	-95.978	POINT (-95.977694 48.99916)	Yes
Laurier	Washington	3016	US-Canada Bord	Jan-24	Trains	4	49	-118.224	POINT (-118.223777 49.00083)	Yes
Laredo	Texas	2304	US-Mexico Bord	Jan-24	Personal Vehicles	425998	27.5	-99.507	POINT (-99.507412 27.499561)	Yes
Sumas	Washington	3009	US-Canada Bord	Jan-24	Buses	14	49.002	-122.265	POINT (-122.264805 49.002388)	Yes
Frontier	Washington	3020	US-Canada Bord	Jan-24	Bus Passengers	194	49	-117.832	POINT (-117.831527 49.0005)	No
Port Huron	Michigan	3802	US-Canada Bord	Jan-24	Truck Containers Empty	9154	42.999	-82.424	POINT (-82.423611 42.998611)	Yes
Pembina	North Dakota	3401	US-Canada Bord	Jan-24	Train Passengers	204	49	-97.237	POINT (-97.237036 49.00453)	No
Fort Fairfield	Maine	107	US-Canada Bord	Jan-24	Trucks	525	46.765	-67.789	POINT (-67.789471 46.765323)	Yes
Alcan	Alaska	3104	US-Canada Bord	Jan-24	Personal Vehicle Pass	1054	62.615	-141.001	POINT (-141.001444 62.614961)	No

Figure1: The original / raw dataset imported from Data.gov.

Port Name	State	Port Code	Border	Date	Measure	Value	Latitude	Longitude	Point	Return
Jackman	Maine	104	US-Canada Border	Jan-24	Trucks	6556	45.806	-70.397	POINT (-70.396722 45.806561)	Yes
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Alcan	Alaska	3104	US-Canada Border	Jan-24	Personal Vehicle Passengers	1054	62.615	-141.001	POINT (-141.001444 62.614961)	No

Figure 2: The cleaned dataset sheet created after data preprocessing.

- This clean dataset sheet was used as the primary source for performing analysis and creating visualizations in the dashboard.

4. Data Preparation:

- Created calculated columns for better analysis (e.g., calculating total border crossing statistics, average rates, etc.).
- This screenshot represent:

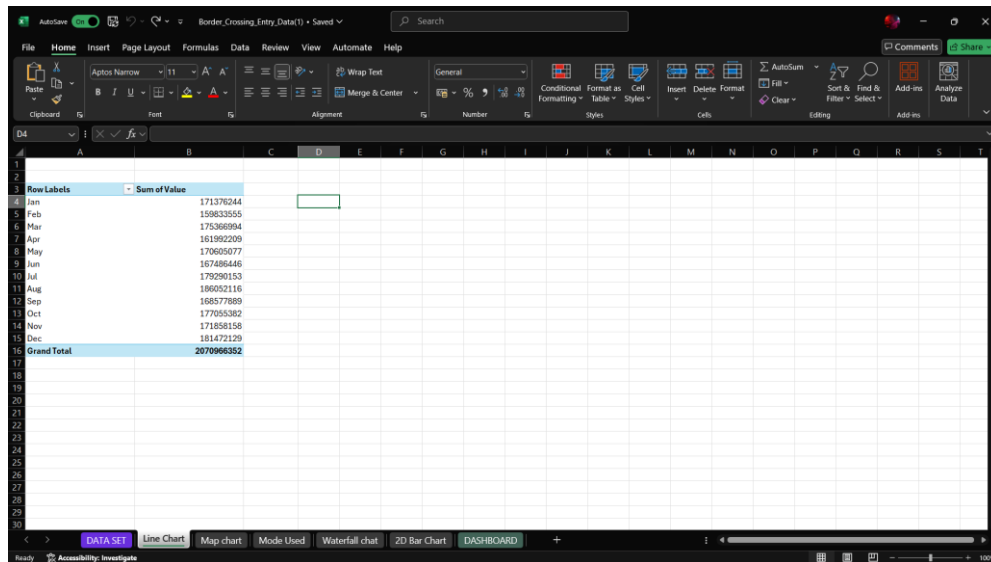


Figure 3: The calculated columns for better analysis

- Used sorting and filtering techniques to analyze specific data points.

This preprocessing phase ensured that the dataset was accurate, consistent, and ready for further analysis using Excel tools like Pivot Tables, Charts, Slicers, and Conditional Formatting.

Chapter 5 - Analysis on dataset

- General Description:

The dataset provides detailed information about border crossing counts across the United States. It covers various aspects and entry point details like gender, race, Hispanic origin, age group, and year-wise data. This comprehensive dataset allows for the observation of patterns and analysis of trends in border crossing statistics over different time periods and across various states.

This project focuses on analysing how border crossing statistics vary between different groups of people. The data helps in identifying which states have higher border crossing statistics, the comparison between male and female border crossing statistics, the most vulnerable age groups, and the differences in border crossing statistics among different races and ethnic groups.

The dataset also enables the study of the changing patterns of border crossing statistics over the years, highlighting any increase or decrease based on several factors. The clean dataset sheet was created to ensure that the data was accurate, consistent, and structured properly for effective analysis. Various Excel tools like Pivot Tables, Charts, Slicers, and Conditional Formatting were used to generate insights and represent them visually through a dynamic dashboard.

This analysis provides meaningful information that can help researchers, policymakers, and healthcare planners understand the border crossing trends in the United States and make data-driven decisions to address this serious issue. The clean dataset was used to generate insights and create various visualizations for the dashboard.

- **Specific Requirements:**

The specific requirements of this project were to analyse the border crossing count dataset and provide meaningful insights using Excel. The project aimed to use the capabilities of Excel to turn raw data into understandable and visually engaging insights through a dashboard. The analysis and dashboard creation were focused on enabling easier decision-making and presenting key data in an effective way.

The main requirements included:

- Cleaning and transforming the raw dataset for accurate and meaningful analysis.
- Creating a separate clean dataset sheet within the Excel workbook for structured analysis and easy accessibility.
- Generating pivot tables to analyse border crossing statistics state-wise, gender-wise, race-wise, and age-wise.
- Designing various charts like bar charts, line charts, pie charts, and heatmaps to visualize different aspects of the data.
- Incorporating slicers to filter the dataset interactively based on parameters like year, state, gender, and age group.
- Highlighting key performance indicators (KPIs) like states with the highest border crossing statistics, lowest border crossing statistics, and average border crossing statistics across the dataset.
- Ensuring the dashboard is dynamic, user-friendly, and provides a comprehensive overview of the data.
- Utilizing conditional formatting to visually emphasize significant values and trends in the data.

These requirements were fulfilled using Excel tools and functions, ensuring an effective analysis of the dataset and the creation of a dynamic and insightful dashboard that can help users better understand border crossing trends in the United States and facilitate data-driven decision-making.

- Analysis results:

Based on the analysis of the border crossing count dataset using Excel, the following key results and insights were obtained:

- State-wise analysis revealed that the border crossing statistics were generally higher among males compared to females across most states and years. This insight highlights the importance of creating border-type specific border management programs to address the issues faced predominantly by the male population.

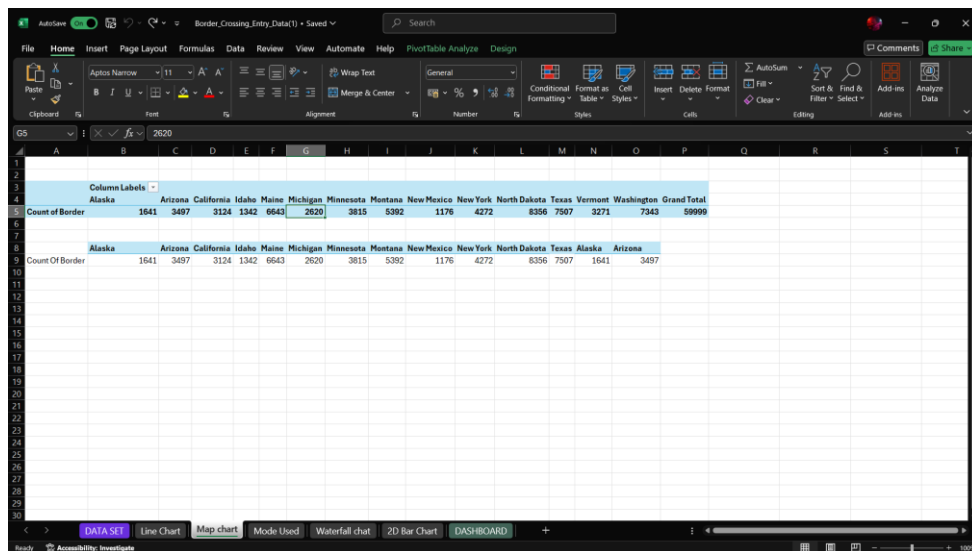


Figure 5: Pivot Tables to analyze border crossing statistics state-wise, gender-wise, race-wise, and age-wise.

- Age-wise analysis highlighted that certain age groups were more vulnerable to border crossing, particularly middle-aged and elderly populations in some regions. The analysis also provided insights into the need for special attention and border management support for younger populations to prevent future increases in border crossing statistics.

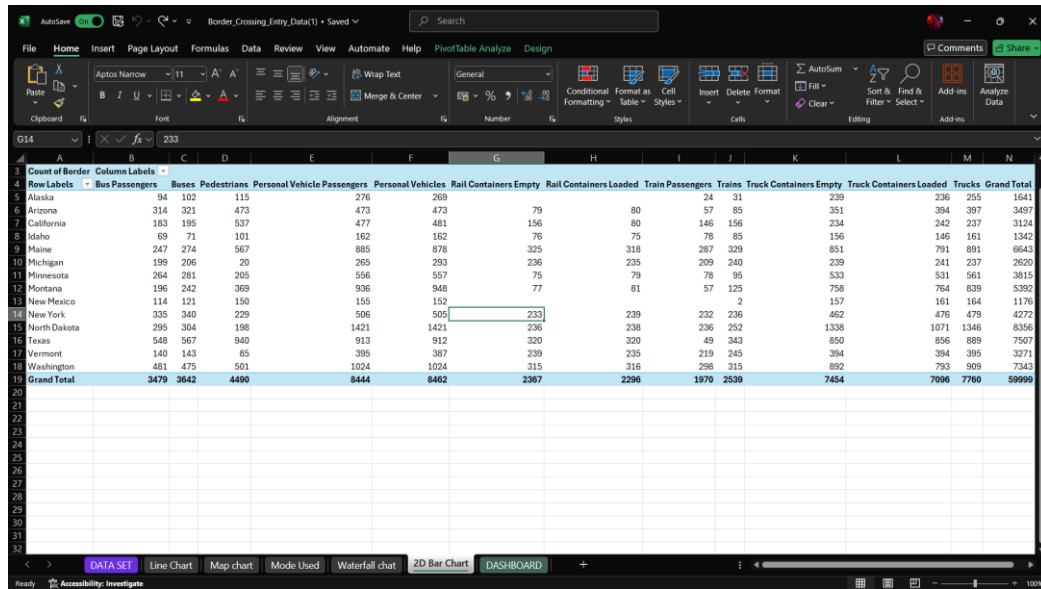


Figure 6: The representation of border crossing statistics

- Race and Hispanic origin analysis helped identify disparities in border crossing statistics among different racial groups, emphasizing the need for targeted border management programs that cater to the unique cultural and social challenges faced by each group.
- Year-wise trend analysis showed variations in border crossing statistics over time, providing valuable insights into whether the rates increased, decreased, or remained stable in specific states or entry point categories. These trends help understand the effectiveness of past border management initiatives and provide a base for planning future programs.

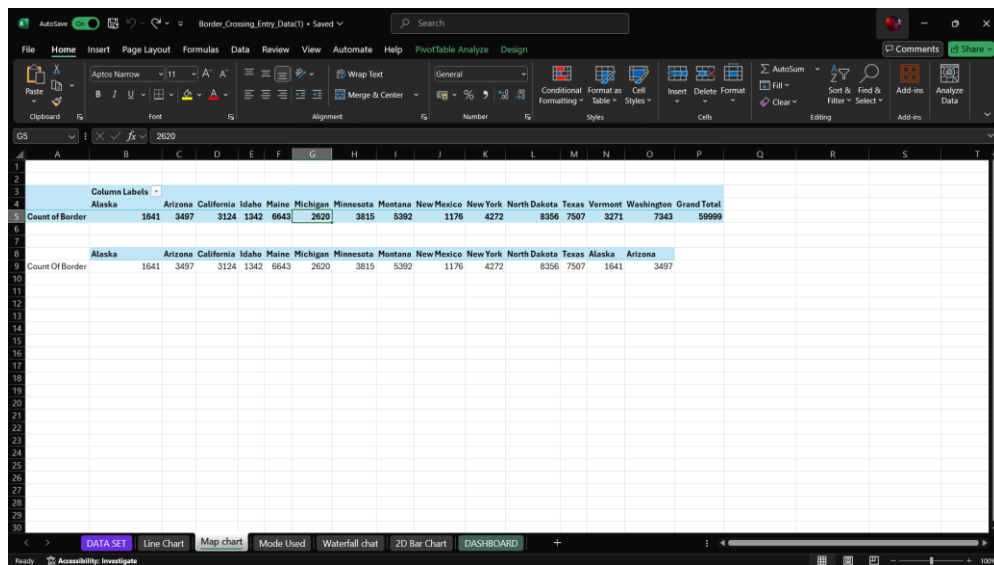


Figure 7: The representation of border crossing

- The use of slicers and KPIs in the dashboard allowed users to interactively filter the data and focus on specific areas of interest, enhancing the overall analytical experience. The interactivity of the dashboard enables users to make quick and informed decisions by analysing different segments of the dataset.

These results provide a clear understanding of border crossing count patterns across the United States and support data-driven decision-making for awareness campaigns and preventive measures. The analysis also showcases the significance of using Excel's powerful features for handling large datasets and converting them into useful insights for practical applications.

- Visualization:

This section presents the visualizations created during the analysis of the Border Crossing dataset using Excel. These visualizations provide a clear and interactive representation of the data for better understanding and decision-making.

The following visualizations were created:

- Bar Chart: Comparison of Average Border Crossing based on Gender

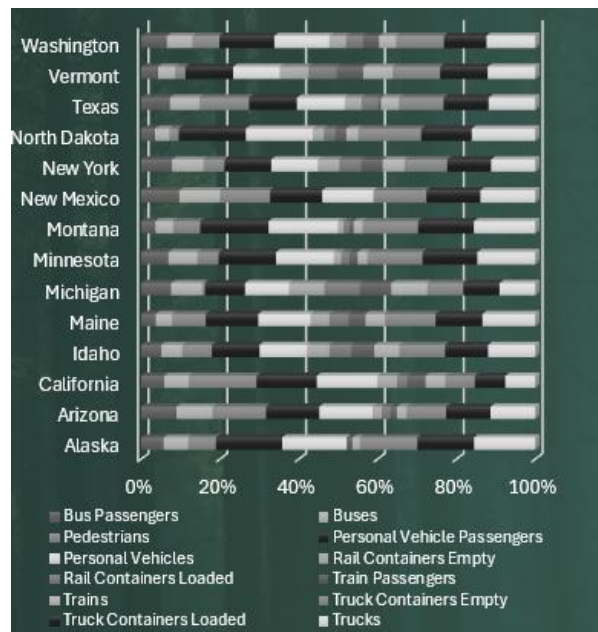


Figure 8: Bar Chart of Average Border Crossing based on Gender

- Pie Chart: Gender-wise Border Crossing Distribution

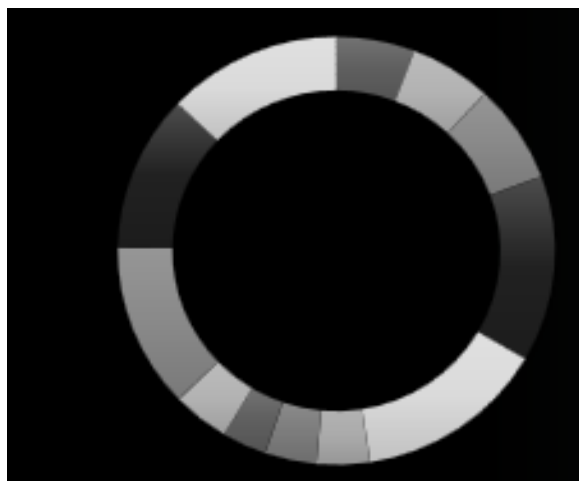


Figure 9: Pie Chart of Gender-wise Border Crossing Distribution

- Line Chart: Year-wise Border Crossing Trend Analysis



Figure 10: Line Chart of Year-wise Border Crossing Trend Analysis

- Water fall Chart: Comparison of Border Crossing across Different Races

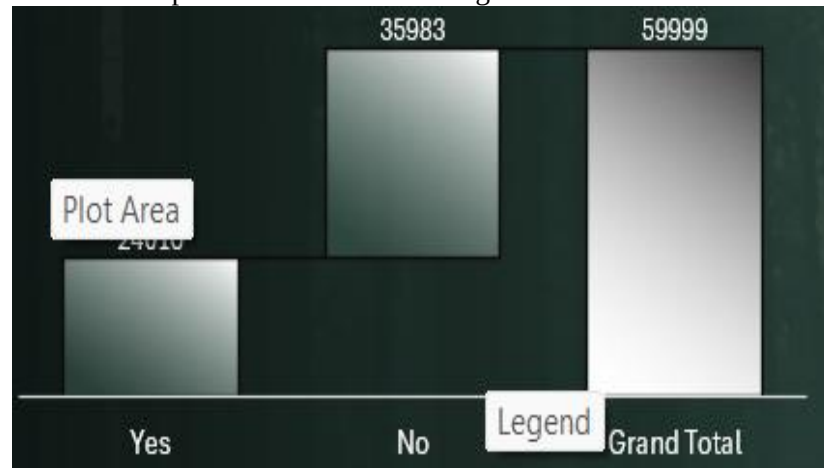


Figure 11: Stack Bar Chart of Comparison of Border Crossing across Different Races

- Map Chart: Distribution of Border Crossing based on Age Groups



Figure 12: Donut Chart of Comparison of Distribution of Border Crossing based on Age Groups

These charts and visuals helped to represent the key insights obtained from the dataset in a visually appealing and user-friendly manner. They allow users to quickly interpret the results and focus on critical areas for better analysis and reporting.

Chapter 6 - Border Crossing Dashboard

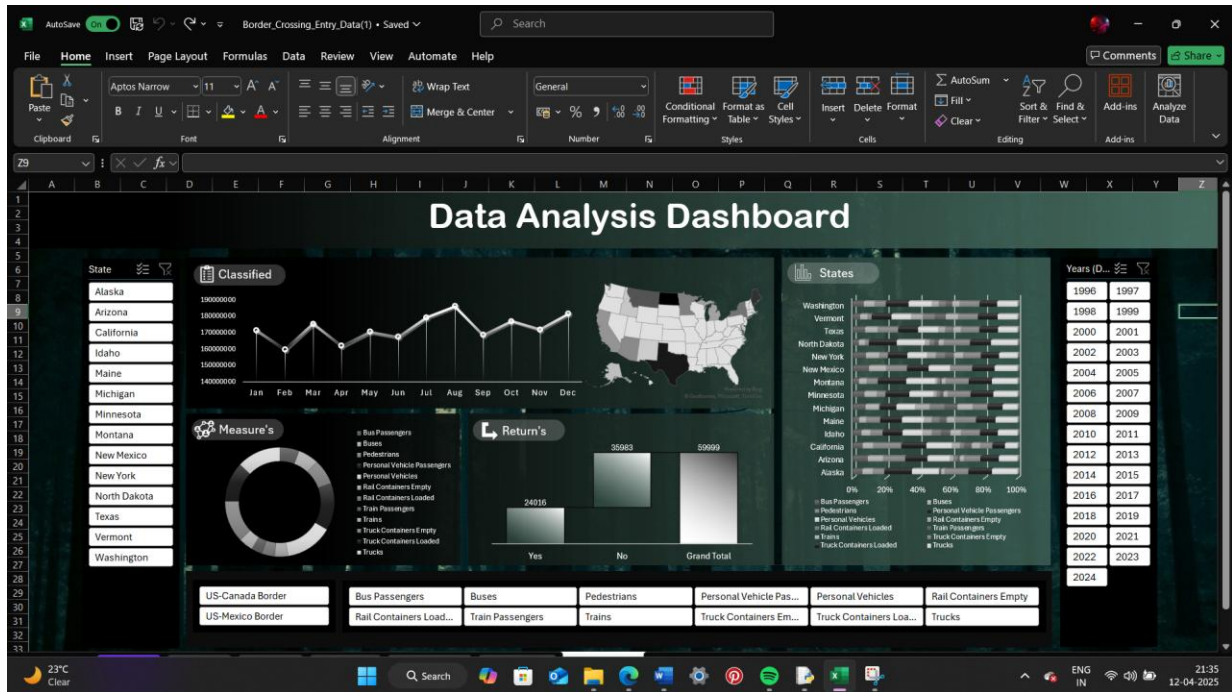


Figure 13: Final Border Crossing Dashboard



Figure 14: Analysing data using slicers



Figure 15: Analysing data using slicers



Figure 16: Analysing data using slicers



Figure 17: Analysing data using slicers



Figure 18: Analysing data using slicers



Figure 19: Analysing data using slicers



Figure 20: Analysing data using slicers



Figure 22: Analysing data using slicers

Chapter 7 - Conclusion

This project provided valuable insights into the border crossing count patterns across different states in the United States. Through the use of Excel tools and techniques, I was able to clean, organize, and analyse the dataset efficiently. The visualizations created using pivot tables and charts helped in representing the data in a user-friendly and interactive way.

The analysis revealed important trends and patterns, such as the higher border crossing statistics among males compared to females, the vulnerability of certain age groups, and the disparities in border crossing statistics among different races. The year-wise trend analysis provided an overview of the changes in border crossing statistics over time, helping in understanding the effectiveness of past health initiatives.

This project also allowed for a detailed examination of how traffic and border parameters influence border crossing statistics. It was identified that specific races and age groups are more susceptible to higher border crossing statistics in certain states. Such findings are essential for the development of focused intervention strategies and for spreading awareness about border management issues.

Overall, this project enhanced my skills in data cleaning, analysis, and visualization using Excel. It demonstrated how large datasets can be transformed into meaningful insights that can aid in decision-making and creating awareness about sensitive social issues like border crossing. It also helped me understand the importance of using data to address real-life problems and the value of presenting data visually to communicate insights effectively.

Working on this project provided me with practical experience in handling real-world data, performing exploratory data analysis, and generating an interactive dashboard to support data-driven decisions. This project not only strengthened my technical skills but also highlighted the societal impact of data analysis in addressing crucial public health challenges.

Chapter 8 - Future scope

This project has successfully demonstrated the ability to use Excel as a powerful tool for data cleaning, analysis, and visualization. However, there is significant scope for future enhancements and further research to expand the effectiveness and applicability of this project to a wider audience and real-world use cases.

In the future, the dataset can be expanded to include more recent data from upcoming years for a more current analysis. Additional parameters such as education level, employment status, socio-economic background, border management history, access to healthcare facilities, and lifestyle factors could be incorporated. Including these parameters would allow for a more detailed and multi-dimensional analysis of the factors contributing to border crossing statistics across different regions.

Furthermore, advanced data visualization tools like Power BI, Tableau, or Python libraries such as Matplotlib and Seaborn can be integrated for creating more dynamic and visually appealing dashboards. These tools can offer better interactivity, deeper analysis, and real-time data visualization capabilities. Implementing these advanced tools would make the analysis more scalable and user-friendly for a wider audience including healthcare professionals, researchers, and government agencies.

In addition, machine learning techniques and predictive modelling could be applied to forecast future border crossing statistics based on historical data trends and entry point patterns. Predictive analytics can enable early identification of high-risk individuals or groups, thereby allowing policymakers to implement targeted prevention strategies effectively.

Another important future enhancement could involve real-time data integration from verified sources such as health agencies, government databases, or border management organizations. This would ensure that the data remains up-to-date, accurate, and relevant for continuous monitoring and proactive analysis.

Collaboration with healthcare providers, educational institutions, and border management organizations can further enhance the utility of this project. These partnerships can assist in developing targeted awareness campaigns, intervention programs, and preventive strategies based on the insights derived from the data analysis.

Thus, this project lays a strong foundation for future developments and has the potential to evolve into a more comprehensive, scalable, and impactful data-driven tool for public health management, border management awareness, and strategic decision-making at a national level.

Chapter 9 - References

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